

# COURSE OVERVIEW

Byeong-hee Roh

Dept. of Software and Computer Engineering

Ajou University



변화와 융합의 중심  
**MMcN**

이동 멀티미디어 융합 네트워크 연구실

Mobile Multimedia convergene Network Lab.

<http://mmcn.ajou.ac.kr>



# Contents

---

- ❖ Network Software : Top-Down Approach
  - TCP/IP Overview
  - Application Layer
    - Standard Socket-based Network Programming
    - Packet Capture Programming
  - Transport/Network Layer
    - Network Kernel Architecture
    - IP Networking with Cisco Packet Tracer
  - Datalink Layer
    - Packet Generation Programming
    - Datalink Layer Implementation Cases: ARP, Switch, etc.
  - Advanced Network Programming – Websocket, IoT
- ❖ Course Overview

# Required Prior-Knowledge

## ❖ Understanding TCP/IP

- **Prerequisite Course: Computer Network**

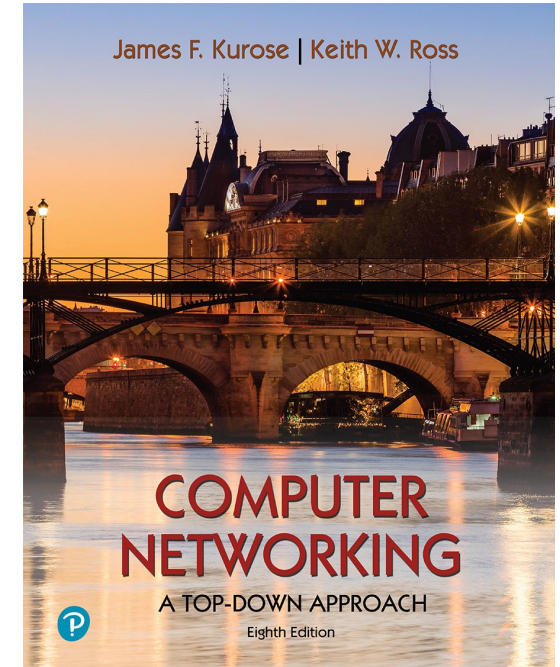
- J. Kurose, K. Ross, Computer Networking :A Top-Down Approach, 8th Edition, Pearson Education, 2022

## ❖ Skills on programming languages

- basically, C using visual studio
- technically, Python, Java Script, Node.js, and other platforms

## ❖ Preparations

- basically, Individual Notebooks
- optionally, shared PCs





# TCP/IP OVERVIEW

# “internet” and “the Internet”

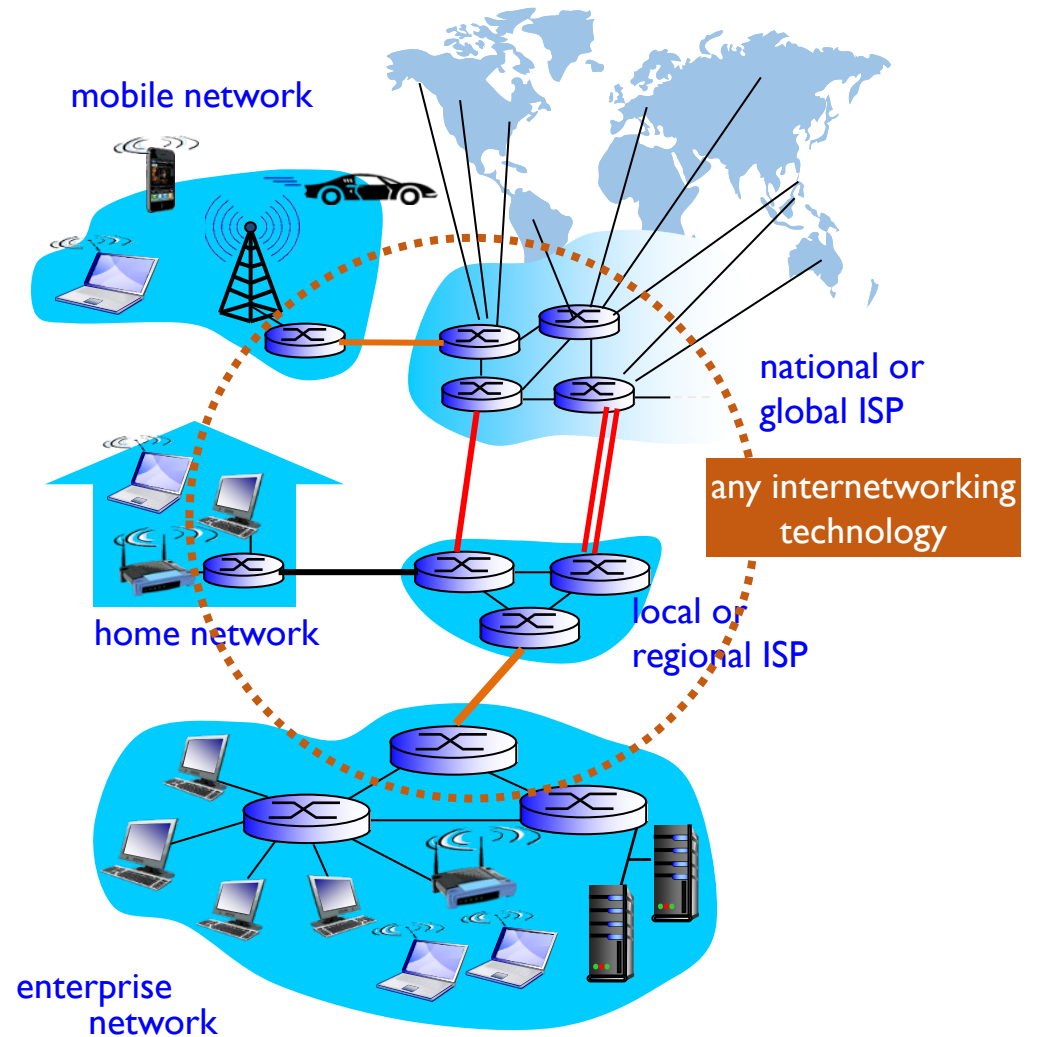
## ❖ internetworking

- technologies to interconnect multiple networks

- X.25
- Frame Relay
- ATM (Asynchronous Transfer Mode)
- etc.

## ❖ internet (as a common noun)

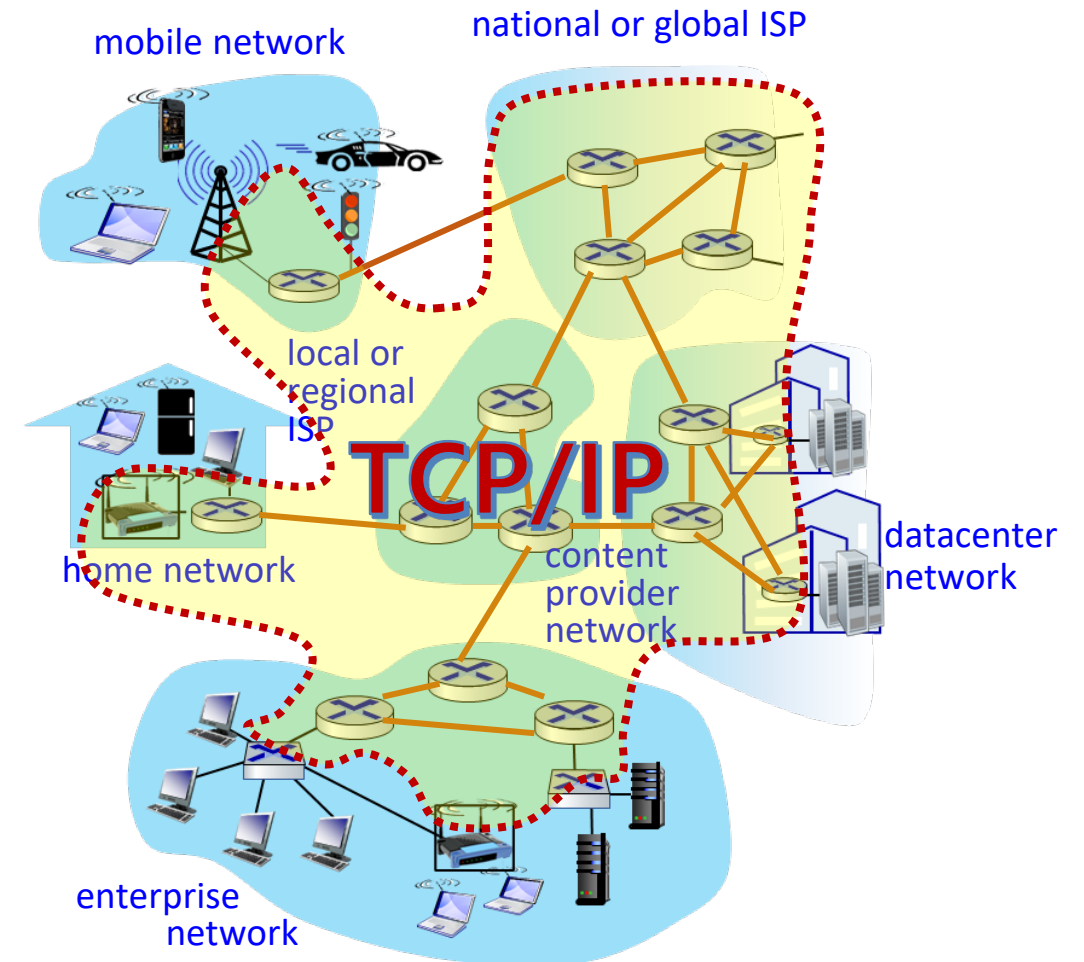
- inter + network(s)
  - any set of inter-connected networks,
  - using various internetworking technologies
- regardless of the use of TCP/IP



# “internet” and “the Internet”

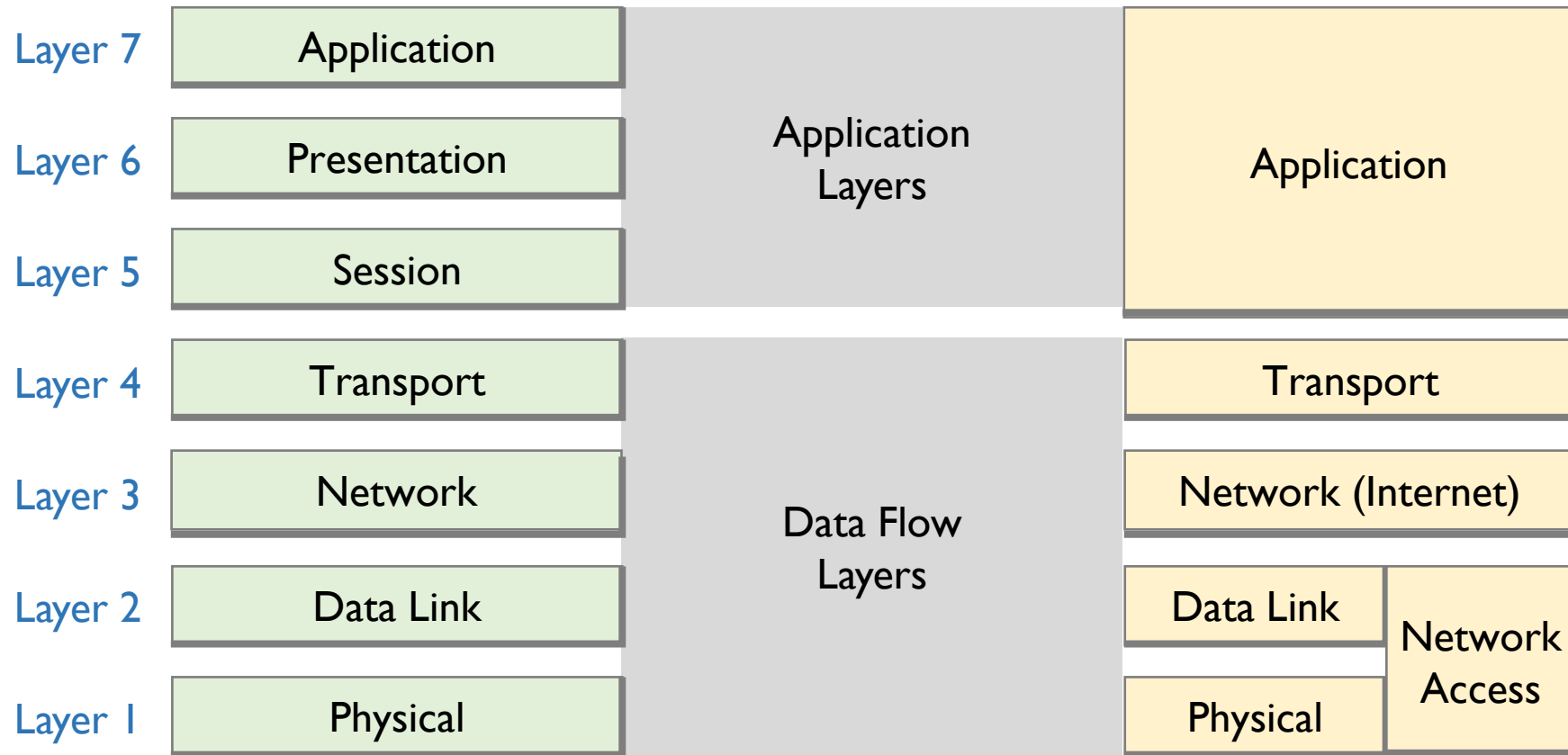
## ❖ the Internet (as a proper noun)

- network of networks
  - network of globally interconnected computer networks
- based on
  - the standard Internet protocol suite
  - TCP/IP
- It consists of millions of
  - connected computing devices,
  - communication links
  - network devices
  - people
  - and everything



# Layered Protocol Reference Model

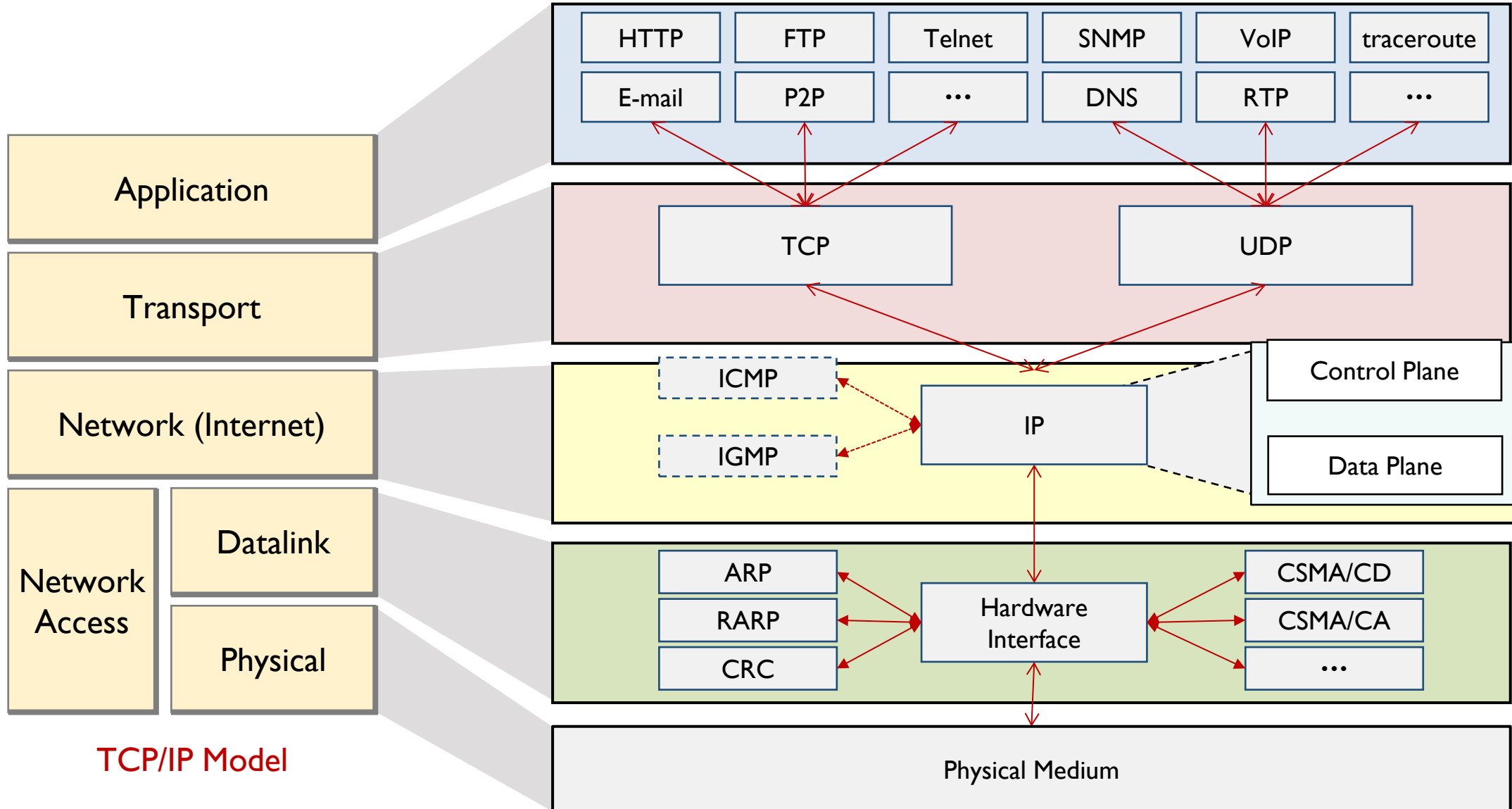
## ❖ OSI and TCP/IP Layered Models



OSI 7 Layers Model

TCP/IP Model

# TCP/IP Protocol Model





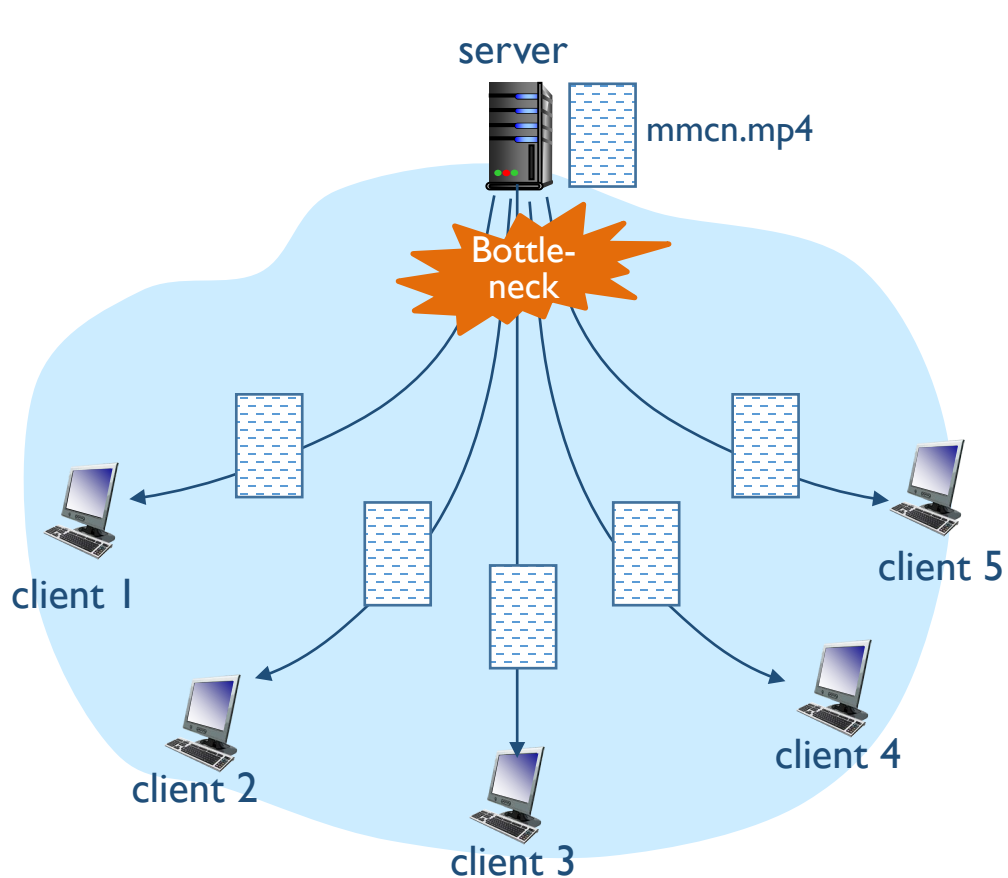


# APPLICATION LAYER

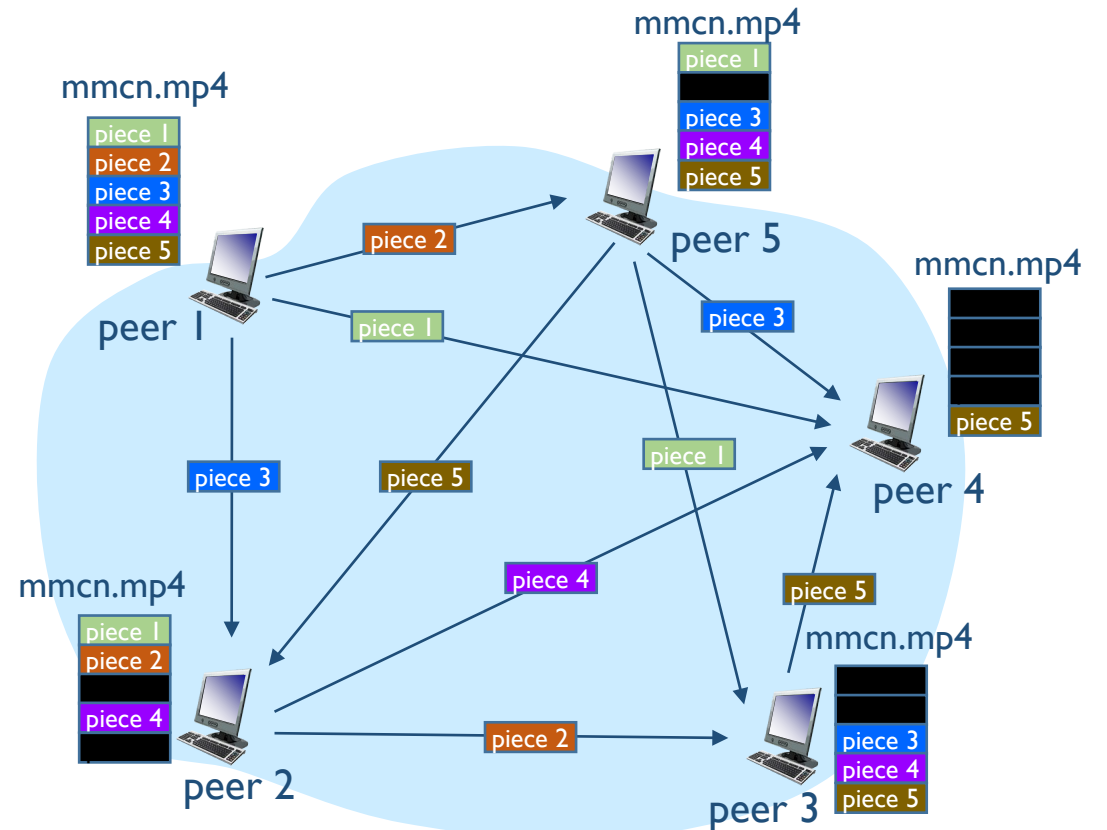
## Network Software: Top-Down Approach

# Application: Architecture

## ❖ Centralized (Client-Server) vs Distributed (Peer-to-Peer)



with a centralized server  
(client-server architecture)

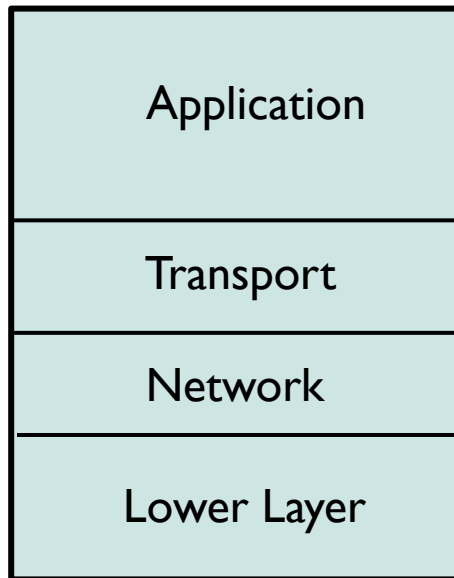


with P2P

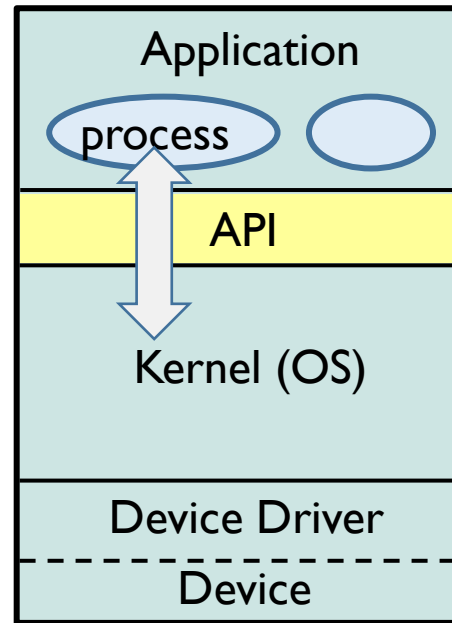
# Application: Socket Basics

## ❖ Layer architectures

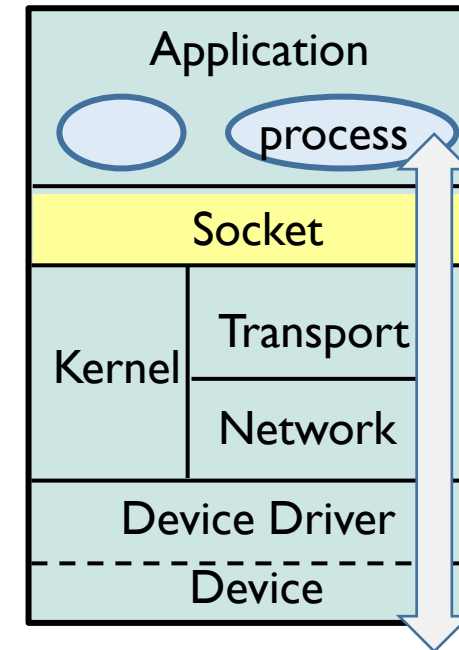
- API – application programming interface
- socket - software interface between process and the network



TCP/IP



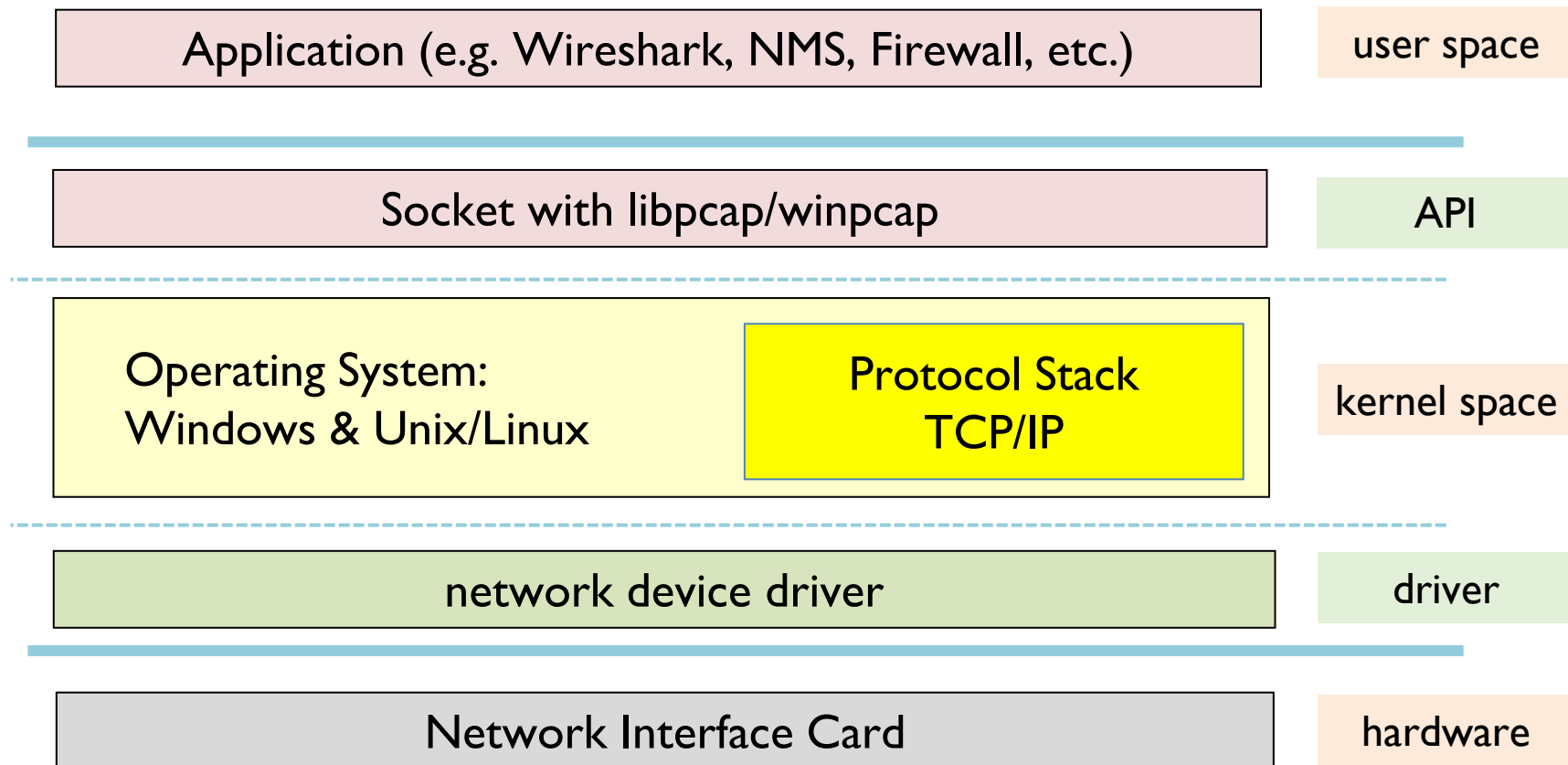
OS



Network Programming

# Application: Socket Basics

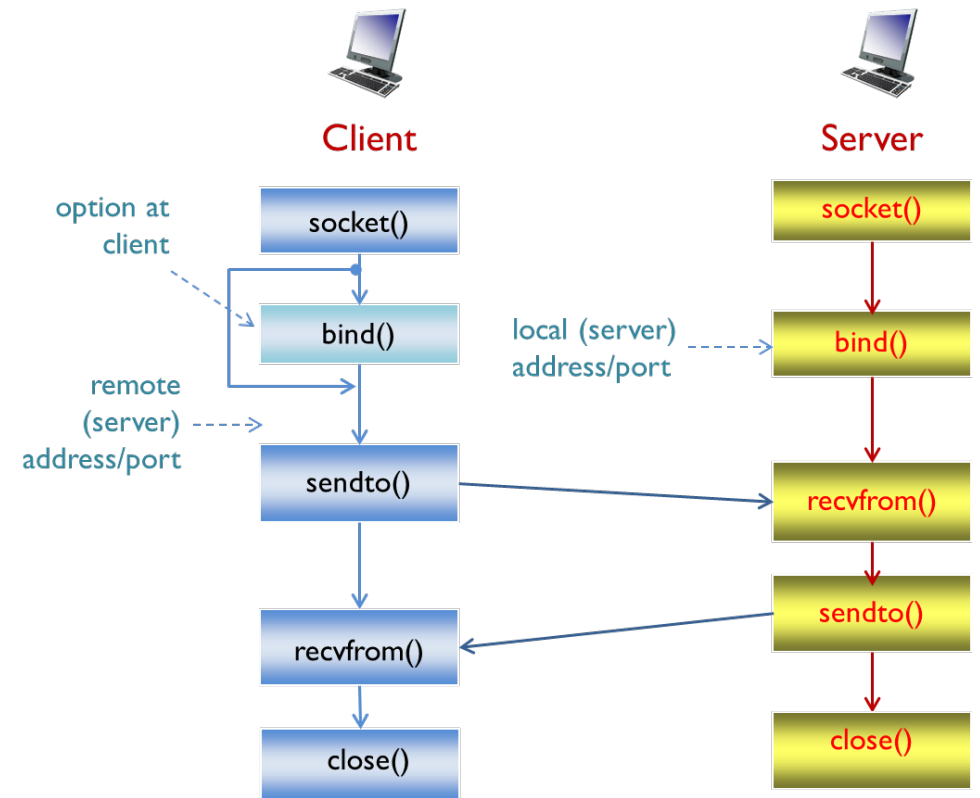
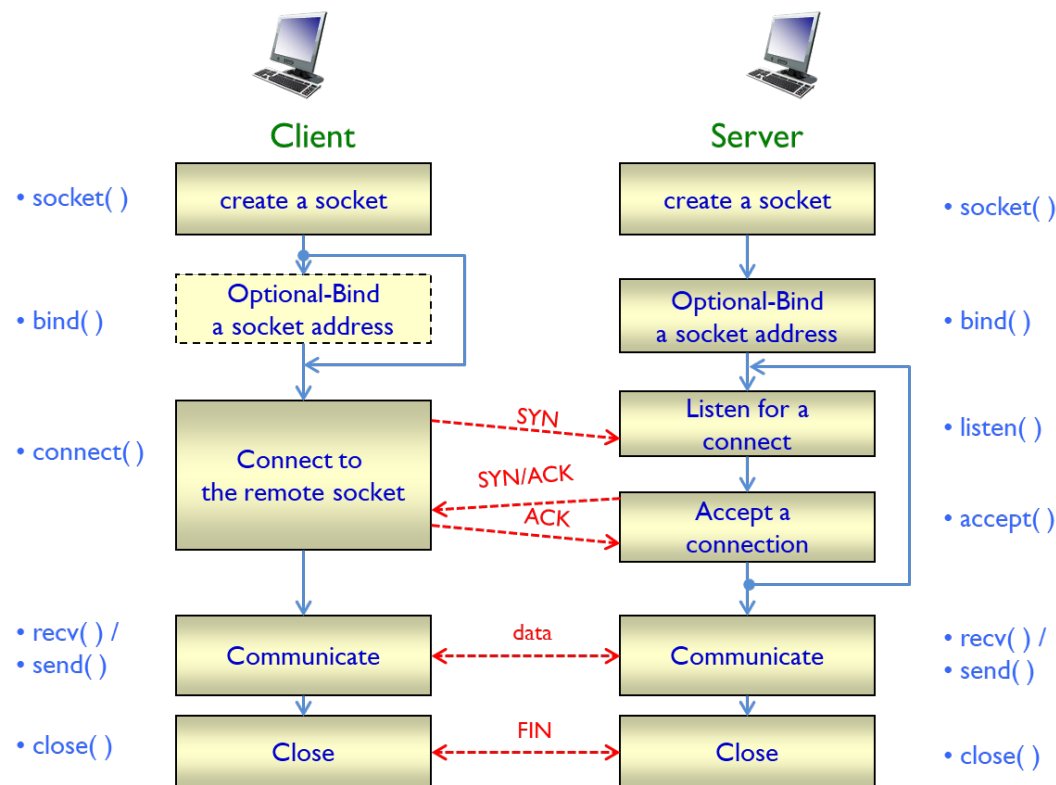
## ❖ Application, OS, Device and Socket



# Application: Standard Socket-based NP

## ❖ Standard Socket Network Programming (NP)

### ▪ TCP and UDP socket programming





# Application: Standard Socket-based NP

---

## ❖ Socket Basics

- Socket Types, Configurations, ...
- Socket options

## ❖ UDP Socket NP

- Connectionless (UDP) programming
- UDP application programming: Broadcast, Multicast, traceroute

## ❖ TCP Socket Network Programming

- Connection Oriented (TCP) programming
- Concurrent Server programming

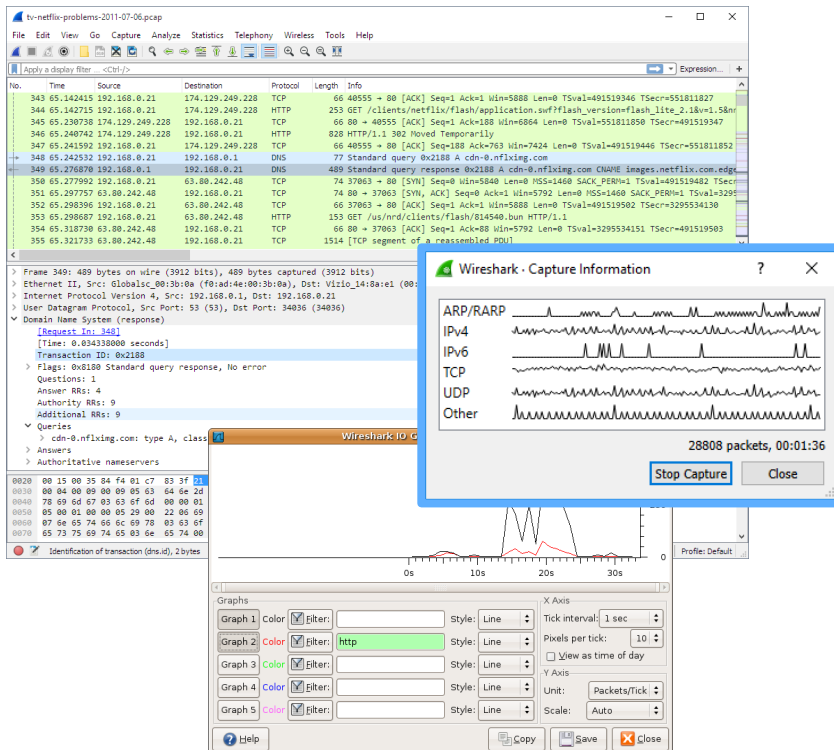
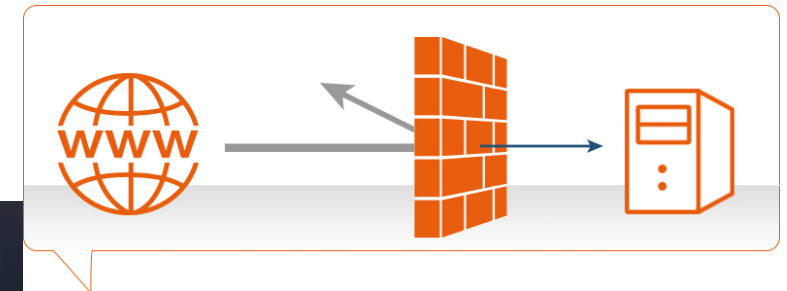
## ❖ Hybrid Socket NP

- Hybrid of TCP and UDP NP

# Application: Packet Capture Programming

## ❖ Network Security and Management

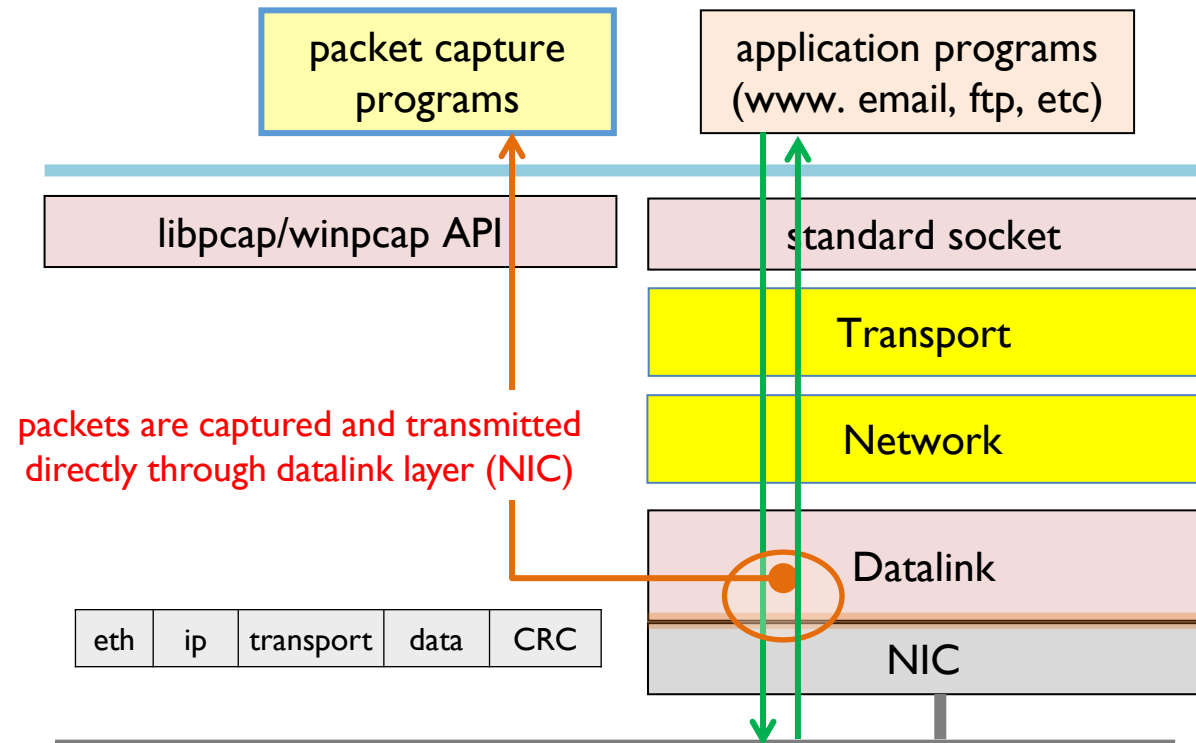
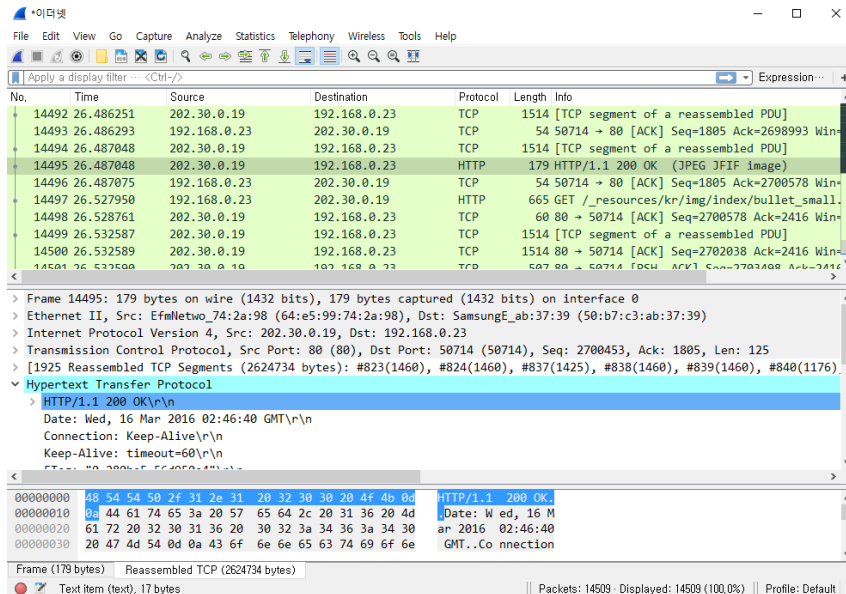
- wireshark - packet capture and analysis tool
- network operation centers
- network security (firewalls, IDS, IPS, ...) devices



# Application: Packet Capture Programming

## ❖ Packet Capture Application Programming

- using pcap library
- packet capture like Wireshark
- network management like SNMP







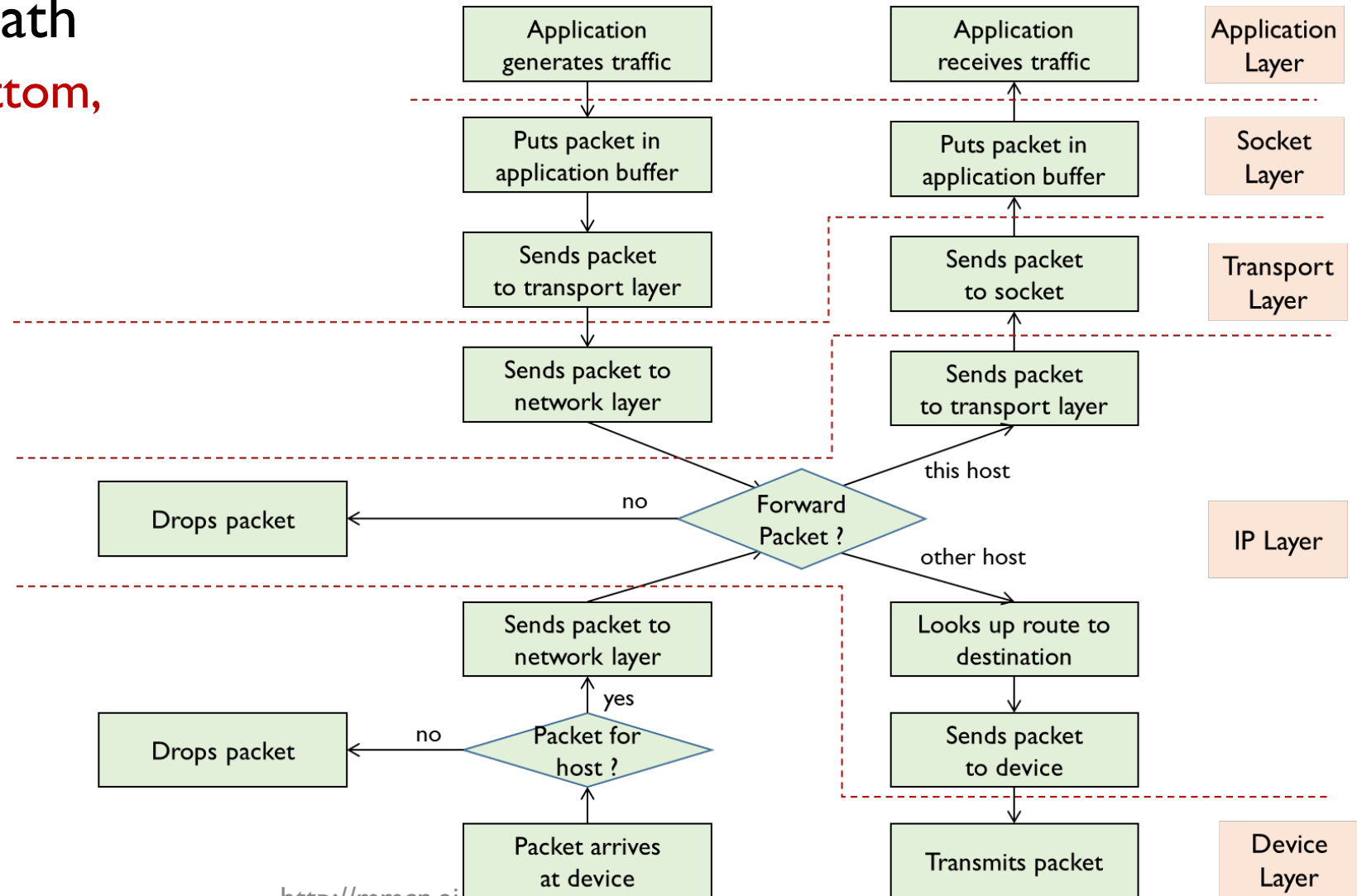
# **TRANSPORT AND IP LAYERS**

## **Network Software: Top-Down Approach**

# From Socket To NIC

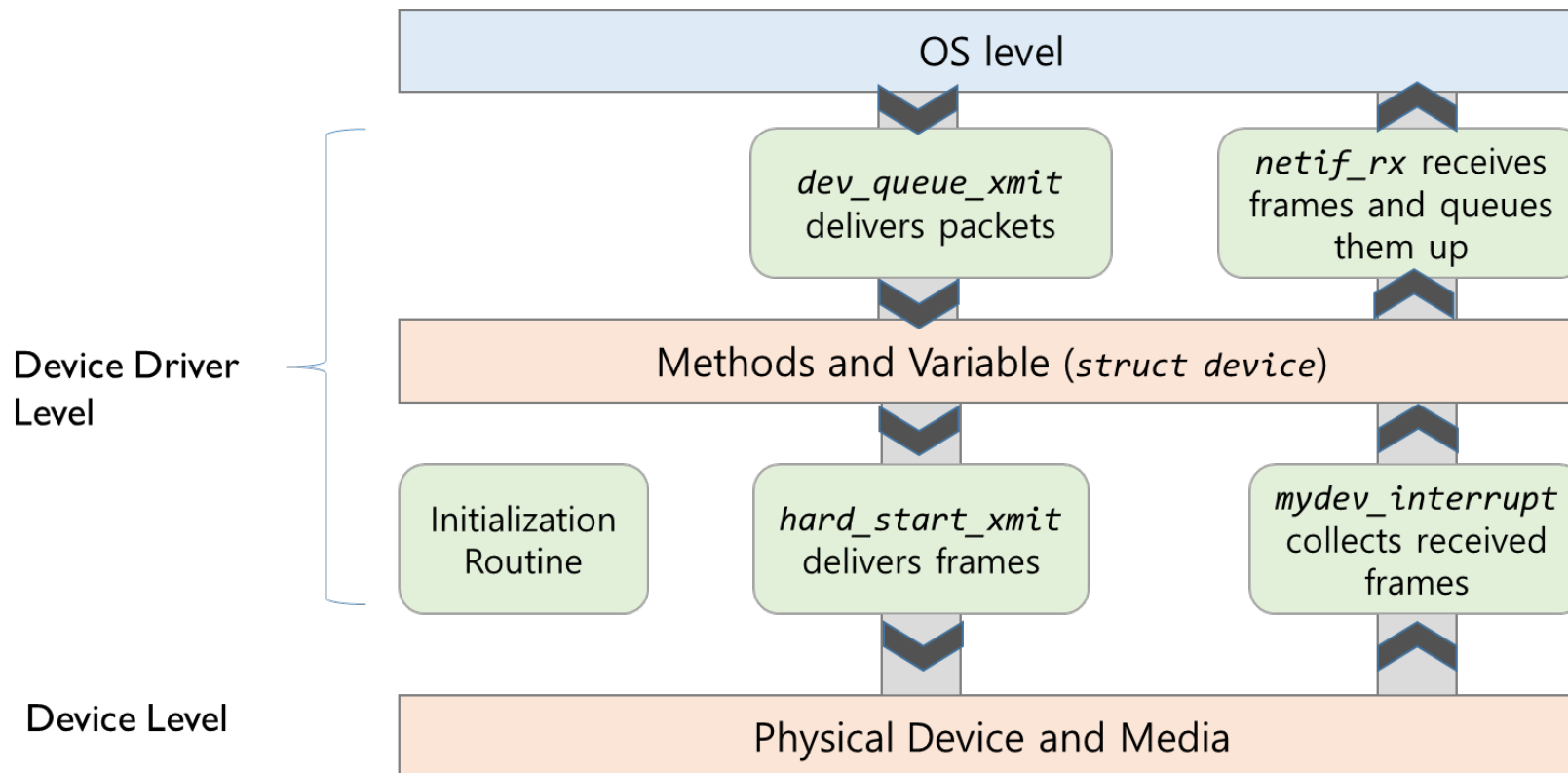
## ❖ Basic Message Path

- from top to bottom,
- and vice versa



# From Socket To NIC

## ❖ Interaction between OS and NICs

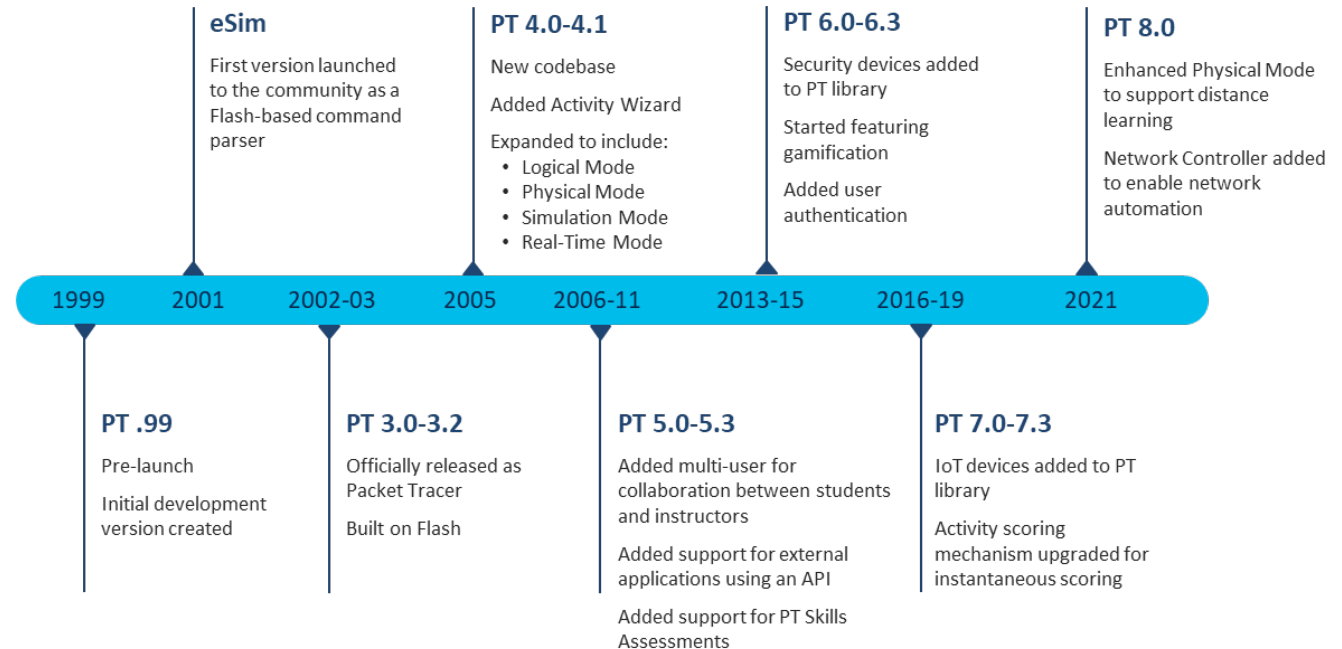
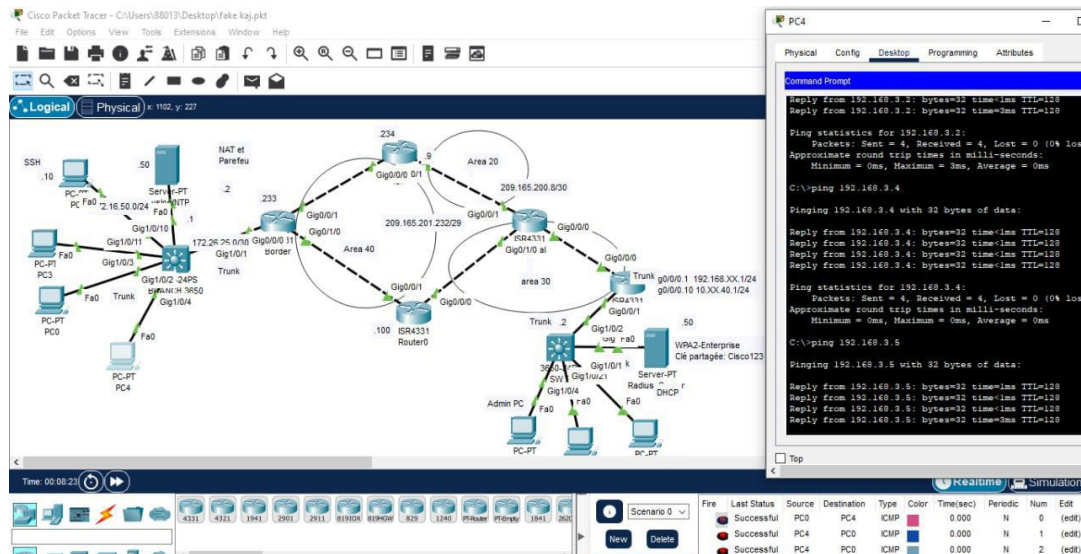


# Networking Configurations

## ❖ Cisco Packet Tracer

### ■ IP Networking

- Network Configuration
- Routing
- and others





# **DATALINK LAYER**

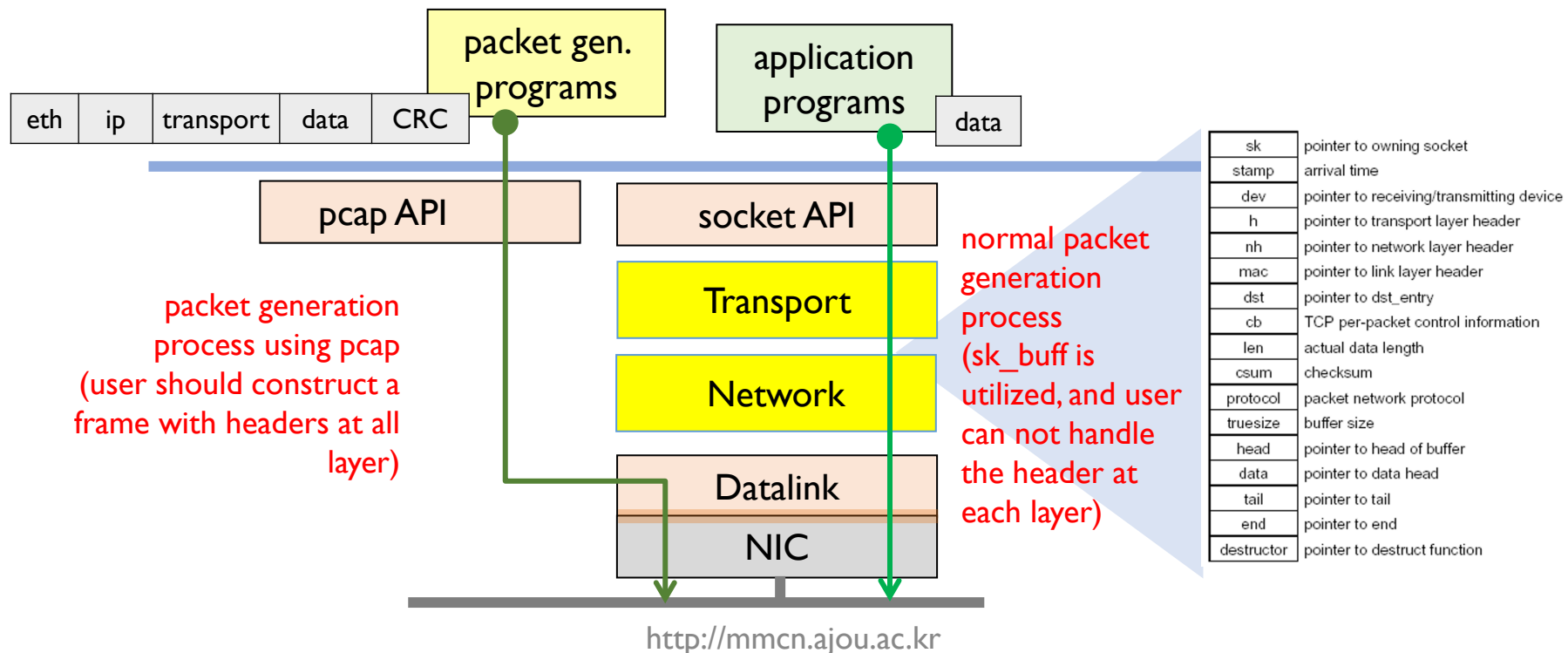
## **Network Software: Top-Down Approach**

# Datalink: Packet Generation Basics

## ❖ pcap API provides packet generation process

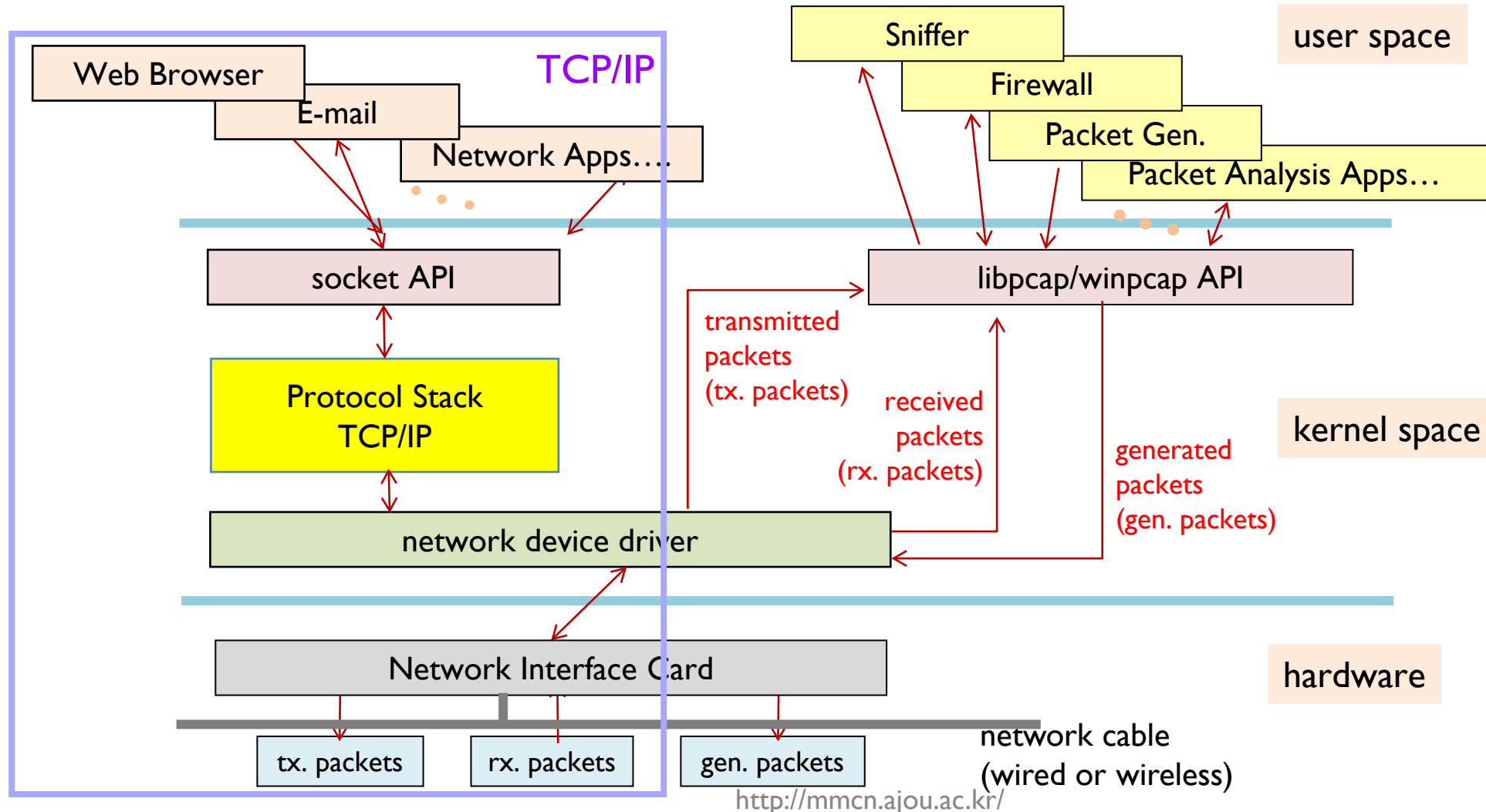
### ▪ an application program

- constructs a full frame format with headers at all layers including application data, then
- sends it through NIC directly using pcap library



# Datalink: Packet Generation Applications

## ❖ Network Programming using both Standard and Raw Sockets





# Datalink: datalink layer functions

---

- ❖ Basics on datalink layer
  - Error Detection (CRC)
  - ARP (Address Resolution Protocol)
  - Ethernet Switch: Self-Learning and Bridging
  - Firewalls at datalink layer





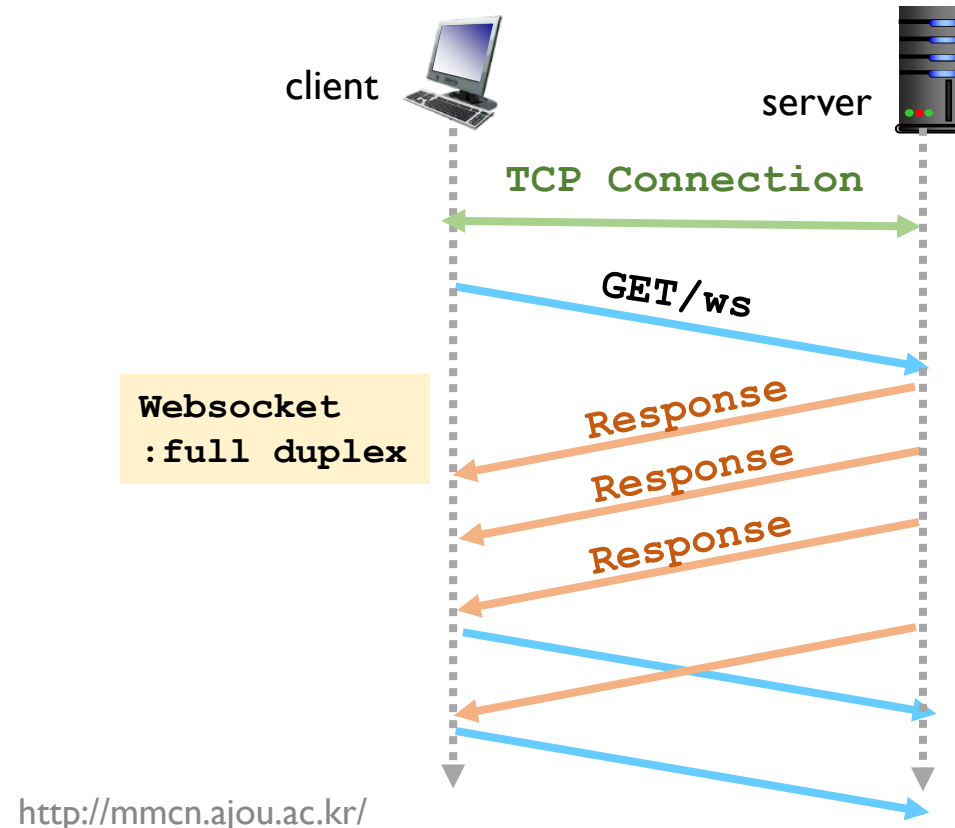
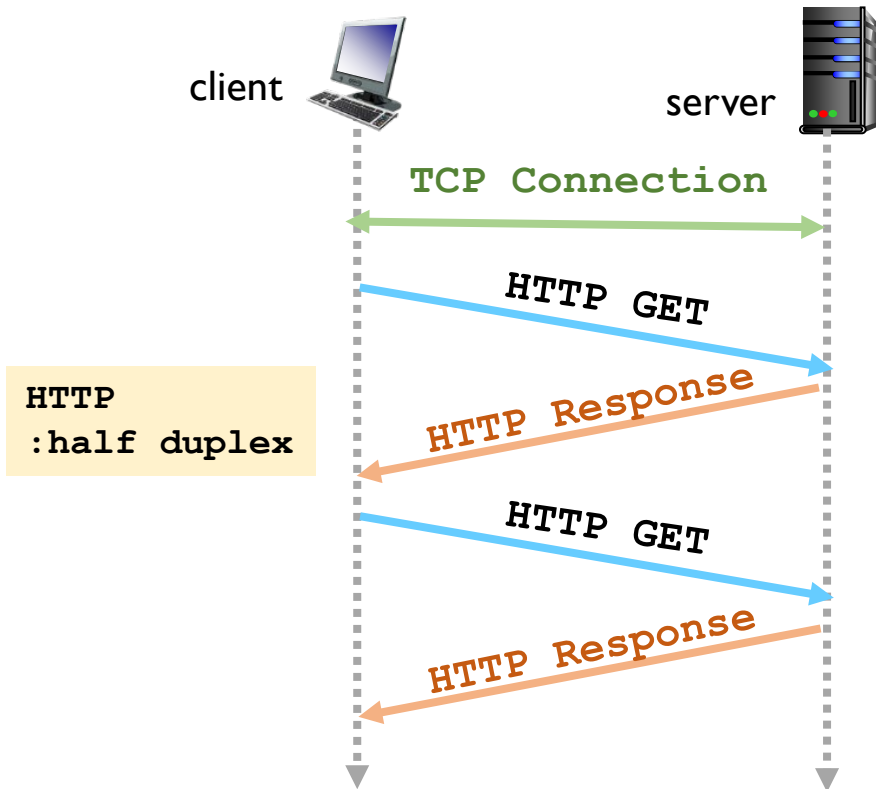
# **ADVANCED TOPICS**

## **Network Software: Top-Down Approach**

# Web Socket

## ❖ Web Socket

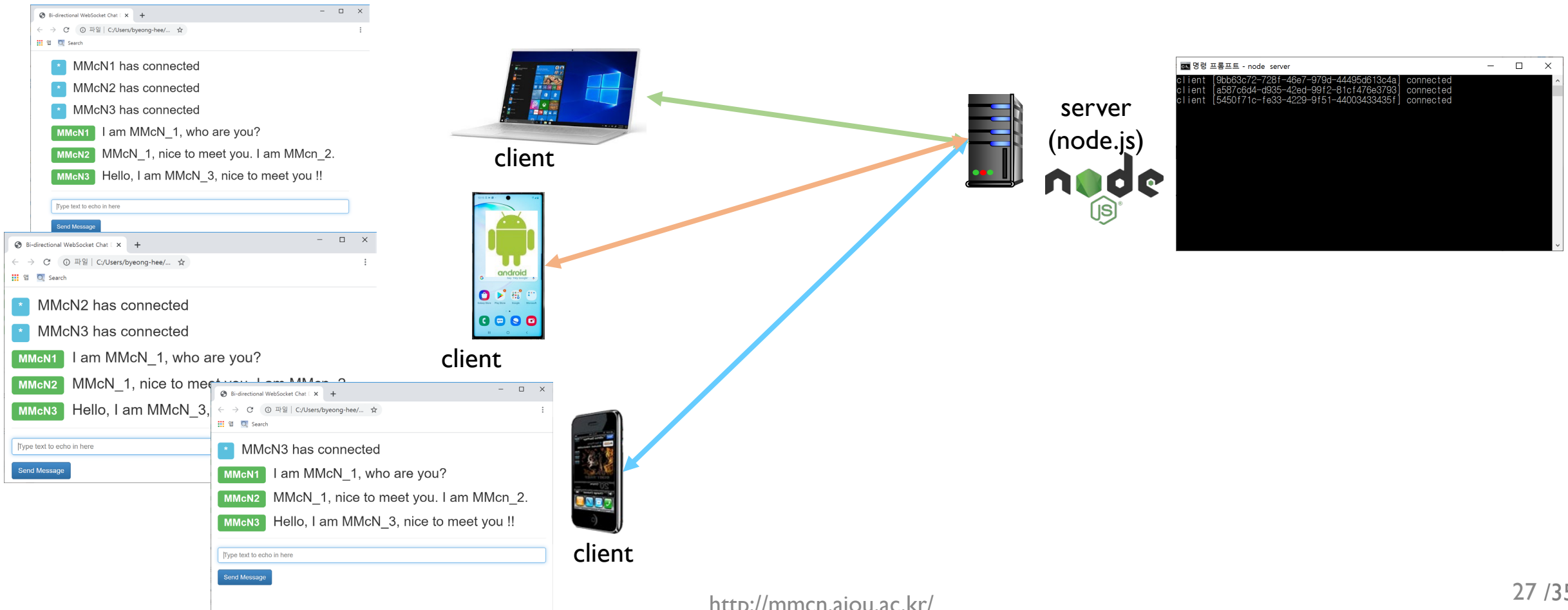
- full-duplex comm. between a client and a server over a TCP connection
- supported in HTML5, work over HTTP port 80 and 443
- RFC 6455, Dec. 2011



# Web Socket

## ❖ Web Socket Programming

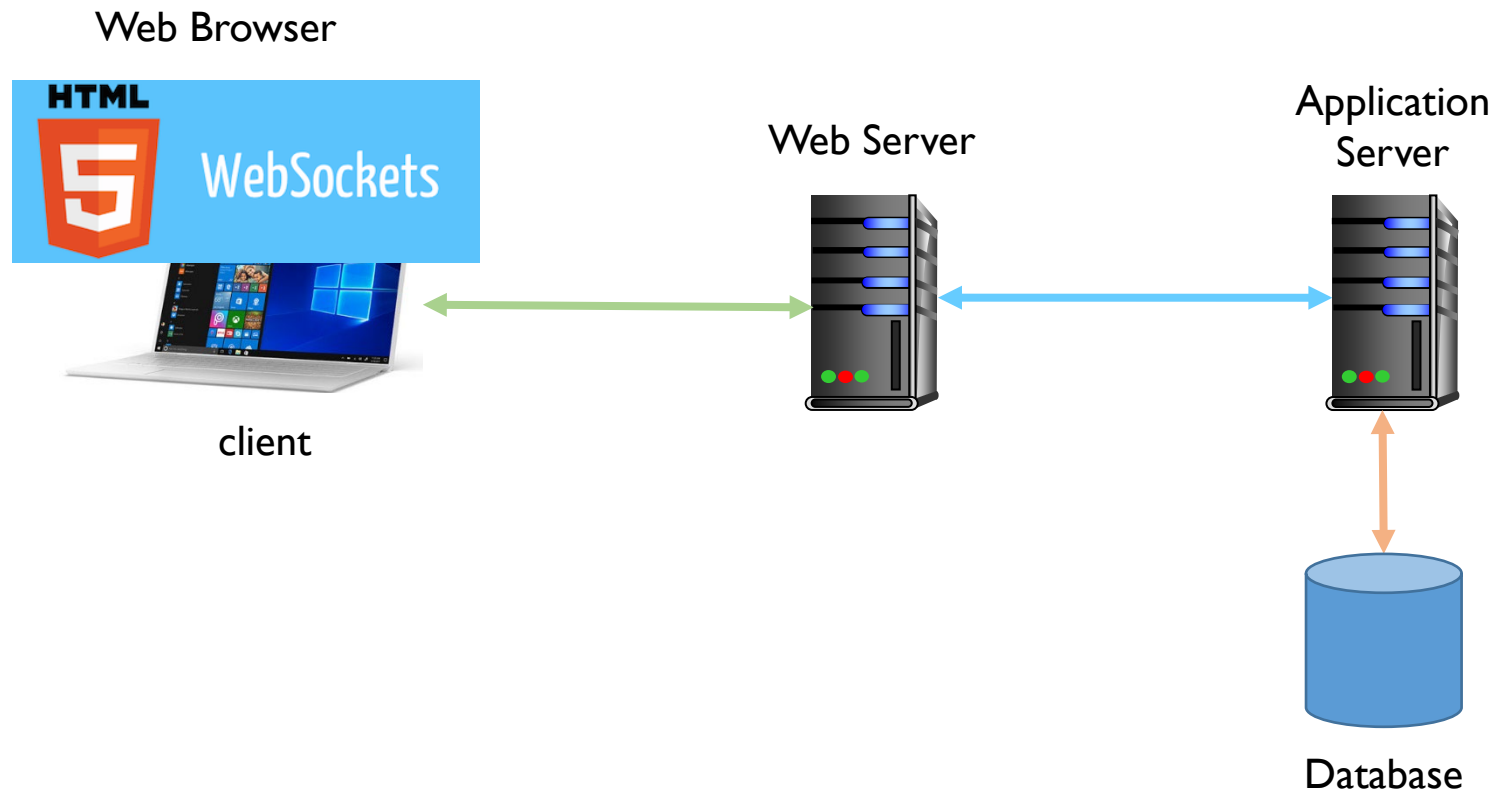
### ▪ chatting programming



# Web-based NP (Optional)

## ❖ Web-based Application Programming

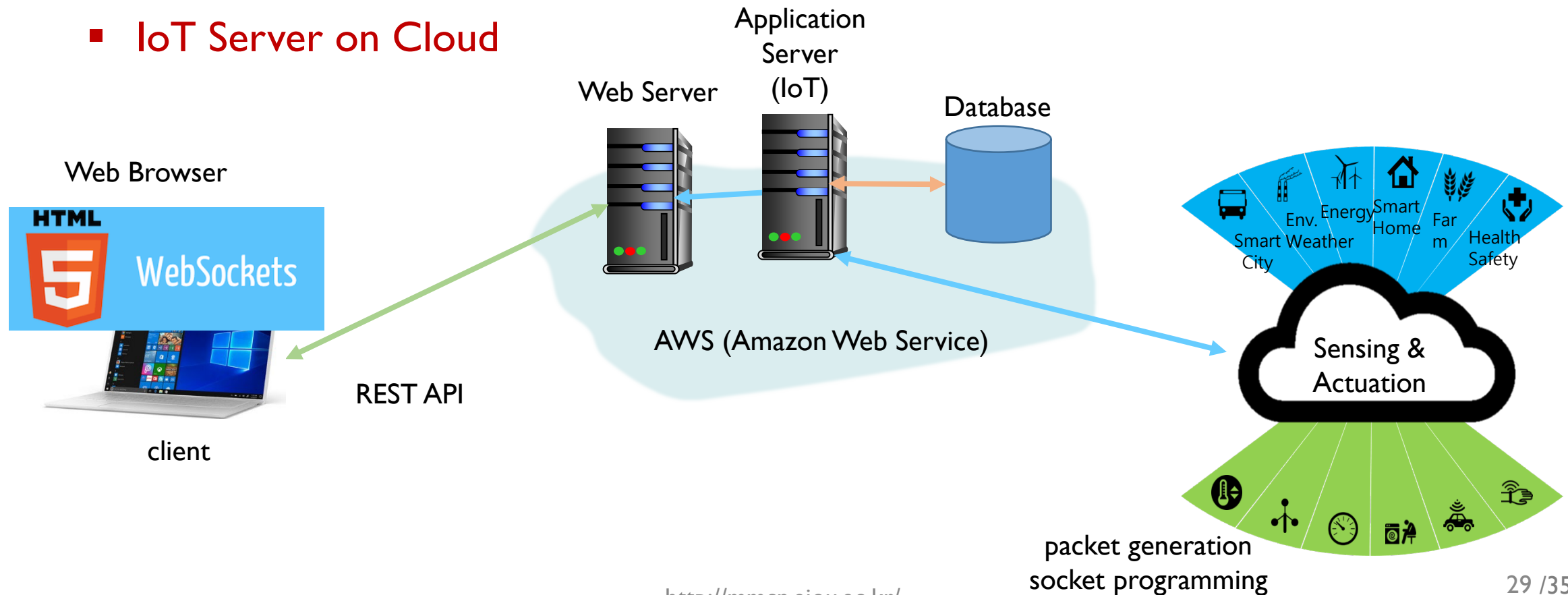
- HTML5 and Web-App programming
- Database Management



# Web-based NP (Optional)

## ❖ Web-based IoT Application Programming

- HTML5 and Web-App programming
- REST API
- IoT Server on Cloud





# COURSE OVERVIEW

---

# 과목 기본 정보개요

## ❖ 과목명

- 네트워크소프트웨어

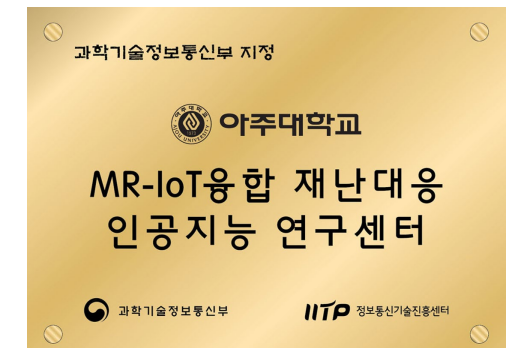
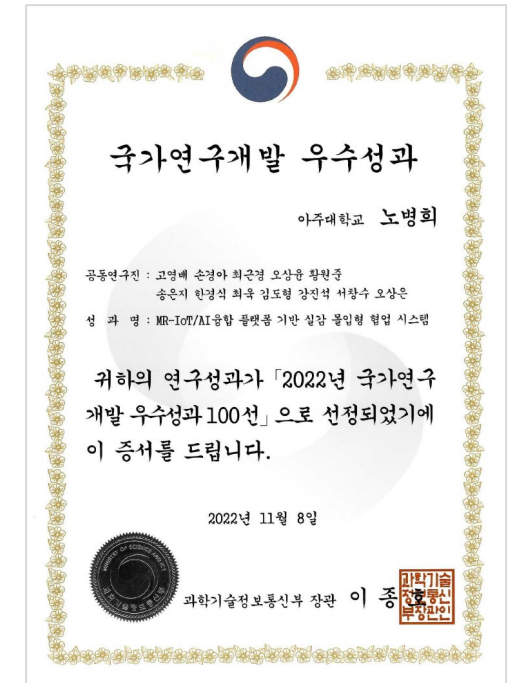
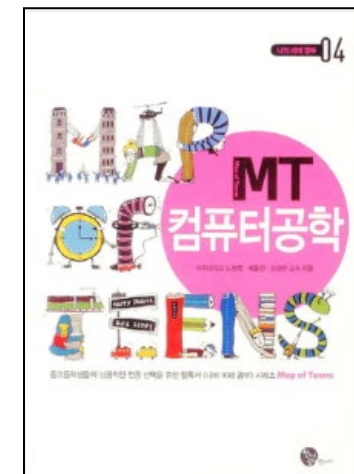
## ❖ 교수 기본 정보

- 담당교수 : 노병희 (Byeong-hee Roh)

- 홈페이지: <http://mmcn.ajou.ac.kr>
- 연구실 : 팔달관 608호 (전화: 031-219-1601)
- 랩실: MMcN 연구실 (팔달관 906호, 2443)
- e-mail: [bhroh\\_at\\_ajou.ac.kr](mailto:bhroh_at_ajou.ac.kr)
- Google Scholar:  
<https://scholar.google.com/citations?user=lquilzcAAAAJ&hl=ko&oi=ao>

## ❖ 과목 기본 정보

- 홈페이지 : [ajouBB](#)
- 조교: TBD





# 과목 목표

---

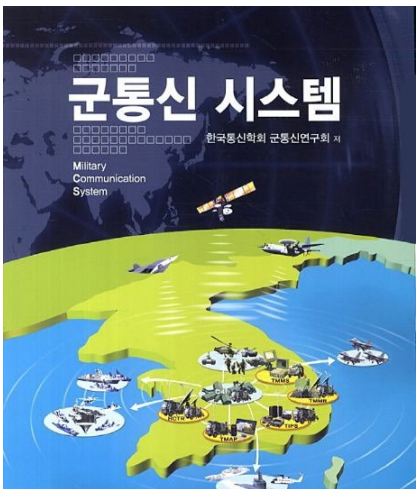
## ❖ 교육 목표

- 이론으로 습득한 TCP/IP 프로토콜 구조에 대한 이해를 증진하고
- 이를 프로그래밍 관점에서 실전 능력을 배양하여,
- 인터넷과 관련한 기술과 응용 서비스의 발전에 기여함.

## ❖ 교육 내용

- TCP/IP Overview
- Application Layer
  - Standard Socket-based Network Programming, Packet Capture Programming
- Transport/Network Layer
  - Network Kernel Architecture, IP Networking with Cisco Packet Tracer
- Datalink Layer
  - Packet Generation Programming, Datalink Layer Implementation Cases: ARP, Switch, etc.
- Advanced Network Programming – Websocket, IoT

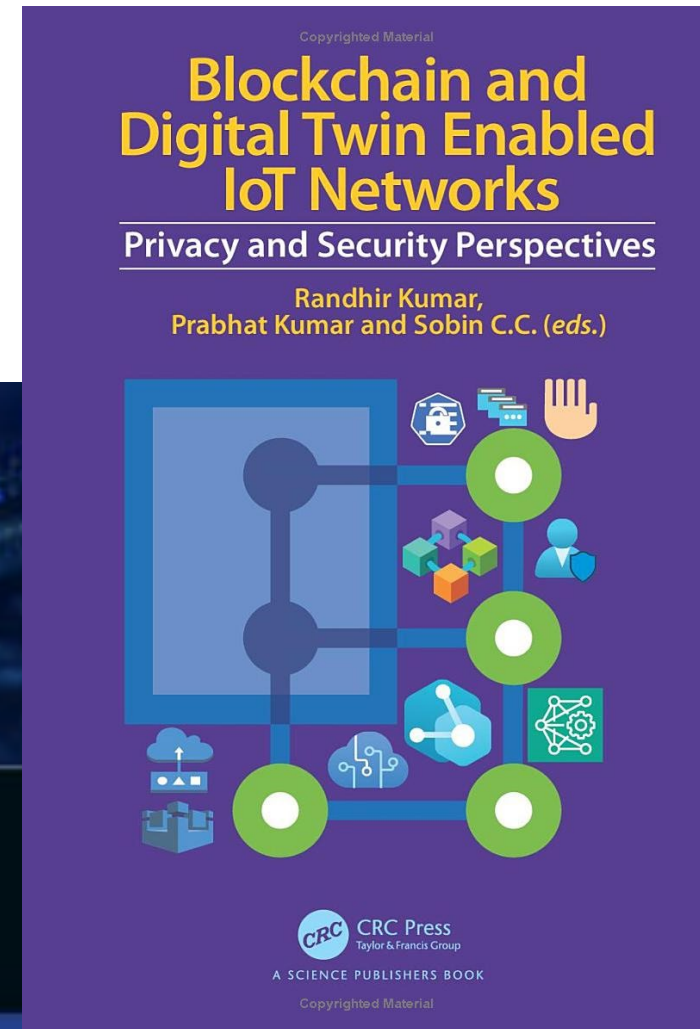
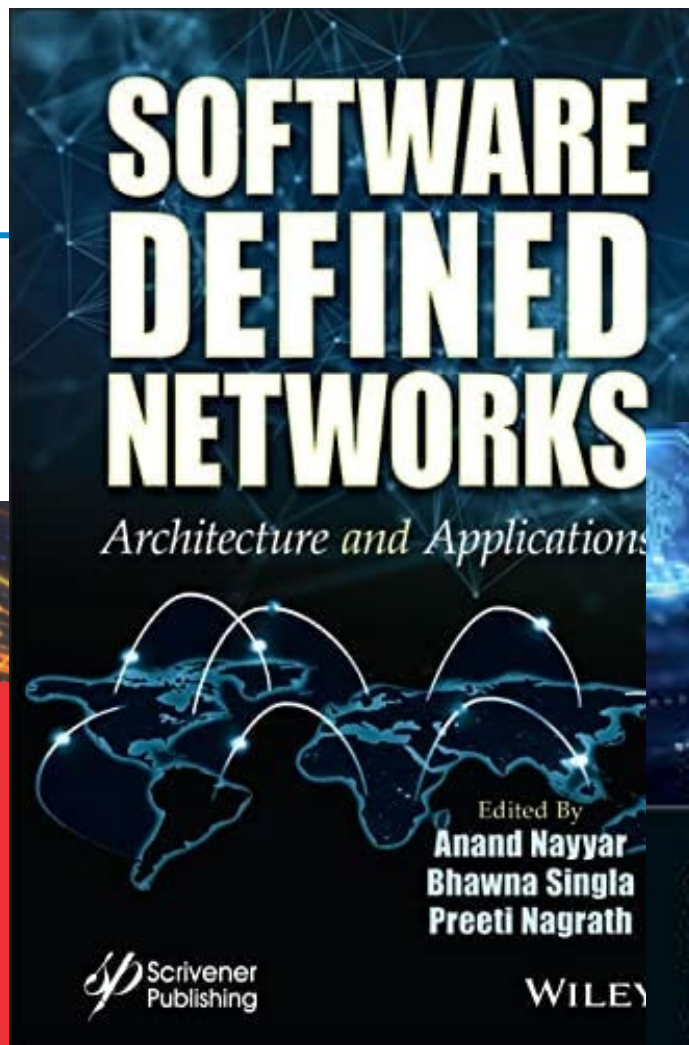




IntechOpen

## Computer-Mediated Communication

*Edited by Indrakshi Dey*



# 과목 개요

- ❖ 학점구성: 이론 (3)
- ❖ 진도계획
  - 01~06주: Application: Socket Programming
  - 07~09주: Application: PCAP Programming + mid exam
  - 10~12주: TCP & IP
  - 14~16주: Datalink and Advanced Topics + final exam
- ❖ 평가방법
  - 중간/기말고사 (30~40%) – 개인 평가
  - 프로그래밍 과제 (40~50%) – 개인 평가
  - 프로젝트 과제 (20%) – 팀 평가
  - 출석: 3회 이후 결석시 1점씩 감점 (공무/경조사 관련 결석은 사유인정)
- ❖ 상대평가2

# Questions ?

MR-IoT/AI Convergence

Future Internet & 5G/6G

Intelligent Networking with AI

MMcN



Mobile **Multimedia Convergence** Network Lab.

Homepage: <http://mmcn.ajou.ac.kr>

facebook: <http://www.facebook.com/#!/groups/190702010947314>

